

CO2 - AT3 -SCENARIO BASED QUESTONS

1. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model.

1. Backpropagation Algorithm in a Multilayer Perceptron (5 Marks)

Definition

Backpropagation is a supervised learning algorithm used to train a neural network. It updates the weights by minimizing the prediction error.

Steps

Step 1: Forward Pass

- Input data is given to the network.
- Each neuron calculates its output using weights and an activation function.
- The network predicts the output.

Example

Input:

- $x_1 = 1$
- $x_2 = 0$

Target Output = 1

Predicted Output = 0.8

Step 2: Error Calculation

Error = Target – Predicted

= 1 – 0.8

= 0.2

Step 3: Backward Propagation

- Error is propagated from the output layer back to the hidden layer.
- Weights are adjusted to reduce the error.

Weight Update Formula:

New Weight = Old Weight + Learning Rate $\times$ Error $\times$ Input

Example:

Old Weight = 0.5

Learning Rate = 0.1

Error = 0.2

Input = 1

New Weight

$= 0.5 + (0.1 \times 0.2 \times 1)$

$= 0.52$

Thus, the weight changes from 0.50 to 0.52.

Role of Learning Rate

- Controls the size of weight updates.
 - Small learning rate → Slow learning but stable.
 - Large learning rate → Faster learning but may overshoot the minimum.
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Role of Activation Function

Examples:

- Sigmoid
- ReLU
- Tanh

Activation functions introduce non-linearity, helping the network learn complex patterns.

Result

Backpropagation reduces prediction error by updating weights repeatedly until the network achieves high classification accuracy.

2. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution

2. Genetic Algorithm (5 Marks)

Dataset

Candidate	Speed	Accuracy	Suitable
A	60	70	No

Candidate	Speed	Accuracy	Suitable
B	80	75	Yes
C	75	85	Yes
D	65	60	No

Assume:

Fitness = Speed + Accuracy

Candidate	Fitness
A	130
B	155
C	160
D	125

Generation 1 (Initial Population)

Best individuals:

- B (155)
- C (160)

These are selected because they have the highest fitness.

Selection

Selection chooses the fittest individuals.

Selected:

- B
- C

Crossover

Parents:

B = (80,75)

C = (75,85)

After crossover:

Child1 = (80,85)

Child2 = (75,75)

Fitness:

Child1 = 165

Child2 = 150

Mutation

Mutate Child1:

Speed changes

80 → 82

Child1 becomes

(82,85)

Fitness
= 167

Generation 2

Candidate	Fitness
Child1	167
Child2	150

Best Solution = Child1

Speed = 82

Accuracy = 85

Fitness = 167

Explanation

Selection

Chooses individuals with the highest fitness.

Crossover

Combines parent characteristics to create better offspring.

Mutation

Randomly changes a gene to increase diversity and avoid local optimum.

Final Optimized Solution

Speed = 82

Accuracy = 85

Fitness = 167

Suitable = Yes

- 3. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs.**

3. Neural Network Classification using Backpropagation (5 Marks)

Sample Dataset

Age	BP	Glucose	Lifestyle Risk	Actual
45	High	High	High	Yes
30	Normal	Normal	Low	No
55	High	High	High	Yes
25	Normal	Normal	Low	No
50	High	Normal	Medium	Yes

Assume the neural network predicts:

Actual	Predicted
Yes	Yes
No	No
Yes	No
No	No
Yes	Yes

Confusion Matrix

- True Positive (TP) = 2
- True Negative (TN) = 2
- False Positive (FP) = 0
- False Negative (FN) = 1

Evaluation Metrics

Accuracy

$$\begin{aligned}\text{Accuracy} &= (TP + TN) / (TP + TN + FP + FN) \\ &= (2 + 2) / 5 \\ &= 80\%\end{aligned}$$

Precision

$$\begin{aligned}\text{Precision} &= TP / (TP + FP) \\ &= 2 / (2 + 0) \\ &= 100\%\end{aligned}$$

Recall

$$\begin{aligned}\text{Recall} &= TP / (TP + FN) \\ &= 2 / (2 + 1) \\ &= 66.7\%\end{aligned}$$

F1-Score

$$\begin{aligned}F1 &= 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \\ &= 2 \times (1 \times 0.667) / (1 + 0.667) \\ &\approx 80\%\end{aligned}$$

How Backpropagation Trains the Network

1. Give patient data as input.
 2. Network predicts disease (Yes/No).
 3. Compare prediction with actual output.
 4. Calculate the error.
 5. Update weights using backpropagation.
 6. Repeat until the error becomes very small.
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Preprocessing

- Remove missing values.
- Normalize Age, BP, and Glucose values.
- Encode categorical values (Yes/No, High/Low) into numbers.

This improves training speed and accuracy.

Feature Selection

Select only important features such as:

- Age
- Blood Pressure
- Glucose Level
- Lifestyle Risk

This reduces unnecessary data, improves accuracy, and decreases training time.

Result

The neural network correctly classifies most patients, achieving 80% accuracy.

Backpropagation improves prediction by continuously updating the network weights, while preprocessing and feature selection increase the model's accuracy and reliability.